Scholar Thesis

Manufacturing Method and Properties of Electric-Car Sliding Bearings Using UHMWPE-Matrix Nano Cu/Graphene Composite

Abstract

Electric vehicles (EVs) demand sliding bearing materials that combine low friction, high wear resistance, and stable performance under electrical and thermal loads. Ultra-high molecular weight polyethylene (UHMWPE) is a promising matrix due to its self-lubricating nature and chemical stability, but its low hardness and poor thermal stability limit its use in demanding tribological contacts. This thesis investigates the manufacturing and properties of UHMWPE-matrix composites reinforced with nano-copper (n-Cu) and graphene for EV sliding bearing applications. A solvent-assisted dispersion and hot-compression molding route is developed to achieve uniform nanofiller distribution. The synergistic effects of n-Cu and graphene on mechanical strength, wear resistance, thermal stability, and electrical conductivity are systematically evaluated. Results demonstrate that hybrid n-Cu/graphene reinforcement significantly reduces wear rate and friction coefficient while enhancing load-bearing capacity. The optimized composite formulation and processing parameters provide a viable pathway for producing high-performance EV sliding bearings.

Keywords: UHMWPE; Nano-copper; Graphene; Sliding bearing; Electric vehicle; Tribological properties

Chapter 1 Introduction

1.1 Research Background

The global transition to electric vehicles has intensified the demand for lightweight, maintenance-free, and durable tribological components. Sliding bearings in EV drivetrains,

steering systems, and thermal management components must operate under conditions of varying load, speed, and temperature while maintaining low friction and minimal wear. The elimination of grease lubrication—driven by sustainability and maintenance-reduction goals—places additional demands on the intrinsic tribological properties of bearing materials.

Ultra-high molecular weight polyethylene (UHMWPE) is an attractive candidate for such applications due to its excellent self-lubrication, high impact resistance, chemical stability, and low water absorption . UHMWPE-based materials have demonstrated exceptional sliding properties and abrasion resistance, with wear loss orders of magnitude lower than POM, PA6, and PBT in ring abrasion tests . However, pure UHMWPE suffers from low surface hardness, poor heat resistance, and inadequate abrasive wear resistance, which compromise its service life under demanding conditions .

Nanofiller reinforcement has emerged as a promising strategy to address these limitations. Copper nanoparticles offer high thermal conductivity and can form protective tribofilms during friction . Graphene, with its exceptional mechanical strength and layered structure, provides efficient load transfer and creates tortuous paths for crack propagation . The combination of metallic and carbon-based nanofillers may produce synergistic effects that neither filler alone can achieve.

1.2 Research Status

1.2.1 UHMWPE/Copper Nanocomposites

Studies on UHMWPE/copper nanocomposites have demonstrated significant improvements in tribological properties. Srivastava et al. prepared UHMWPE/HDPE/copper nanocomposites by melt mixing with copper contents of 0.1%, 0.5%, 1.0%, and 5.0%, evaluating microstructure, tribological, thermal, and mechanical properties . Inclusion of small amounts (0.1–0.5 wt%) of copper nanoparticles resulted in significant improvement in sliding wear performance .

The mechanism of wear reduction involves the formation of protective tribofilms. Vasilev et al. demonstrated that UHMWPE nanocomposites filled with 2 wt% CuO/MBT exhibited a ~3.2 times reduction in wear rate compared to pure polymer, with the formation of secondary

structures in the form of tribofilms protecting the material from wear . The wear mechanism transformed from adhesive to fatigue wear.

For copper nanowires in polyethylene matrices, oleylamine coating is essential to achieve dispersion. Copper nanowires with aspect ratios on the order of 1000 can be randomly dispersed in polyethylene when coated with oleylamine, with agglomeration almost completely suppressed . Silver nanosheet-coated copper nanowires have been shown to effectively increase thermal stability and wear resistance of UHMWPE composites .

1.2.2 UHMWPE/Graphene Nanocomposites

Graphene and graphene oxide (GO) have been extensively studied as reinforcements for UHMWPE. The addition of GO reduces the wear rate of UHMWPE under both water and seawater lubrication conditions, with wear rate gradually decreasing with increasing GO content .

Graphene spatial orientation significantly affects tribological performance. UHMWPE filled with graphene oriented at 90° exhibited a wear rate reduction of 97.29% compared to pure UHMWPE, along with good thermal conductivity and stable structure . Molecular dynamics simulations have shown that functionalized graphene with hydroxyl, carboxyl, and ester groups increases Young's modulus, bulk modulus, and shear modulus of rubber composites, while reducing friction coefficient .

Hybrid GO/MWCNT-COOH fillers in UHMWPE have demonstrated synergistic enhancement. The hybrid U-GO-CNT composite consistently demonstrated the most pronounced improvements in thermal stability, surface wettability, and mechanical properties, attributed to the formation of an interconnected 3D hybrid network where 1D MWCNTs bridge 2D GO sheets, promoting efficient stress transfer and creating a formidable barrier to crack propagation .

1.3 Research Objectives

This thesis aims to develop and characterize UHMWPE-matrix nano Cu/graphene composites for electric-car sliding bearing applications. Specific objectives include:

(1) Developing a manufacturing route for uniform dispersion of n-Cu and graphene in UHMWPE;

(2) Optimizing filler content and ratio for balanced mechanical and tribological properties;

(3) Evaluating the synergistic effects of hybrid n-Cu/graphene reinforcement;

(4) Characterizing wear mechanisms and tribofilm formation;

(5) Assessing the suitability of the developed composites for EV sliding bearing applications.

Chapter 2 Experimental Methods

2.1 Materials

UHMWPE powder (GUR 4120, molecular weight ~5×10^6 g/mol) was obtained from Ticona. Copper nanoparticles (n-Cu) with average particle size of 25–60 nm were synthesized by polyol reduction method . Graphene nanoplatelets (GNPs) with thickness of 1–10 nm and diameter of 100 nm to several micrometers were used. Oleylamine (80–90%) and 1-octadecene were used as received.

2.2 Synthesis of Copper Nanoparticles

Copper nanoparticles were synthesized via a double-salt reduction method in ethylene glycol. Copper nitrate and copper sulfate were reduced in the presence of polyvinylpyrrolidone (PVP) as a dispersant . The uniform size of spherical copper nanoparticles was controlled within the range of 10–55 nm by varying synthesis parameters. For improved oxidation resistance, a Cu-CuO core-shell structure was also prepared.

2.3 Surface Modification of Fillers

Copper nanoparticles were coated with oleylamine to improve compatibility with the non-polar UHMWPE matrix . The coating process involved dispersing Cu nanoparticles in a solution containing oleylamine, followed by washing and drying. Graphene was used as-received or functionalized with carboxyl groups to enhance interfacial bonding.

2.4 Composite Preparation

UHMWPE composites containing n-Cu (0.1, 0.5, 1.0, 2.0 wt%) and graphene (0.5, 1.0, 2.0 wt%), both individually and in combination, were prepared by solvent-assisted mixing followed by hot compression molding. The process involved:

1. Ultrasonic dispersion of nanofillers in ethanol
2. Mixing with UHMWPE powder
3. Drying and ball milling for homogeneous distribution
4. Hot compression at 190°C and 10 MPa pressure

2.5 Characterization Methods

Scanning Electron Microscopy (SEM): Fracture surfaces and worn surfaces were examined using SEM to evaluate filler dispersion and wear mechanisms.

X-ray Diffraction (XRD): Crystallinity and phase composition were analyzed.

Differential Scanning Calorimetry (DSC): Melting behavior and crystallinity were determined.

Thermogravimetric Analysis (TGA): Thermal stability was evaluated.

Tribological Testing: Pin-on-disc tests were conducted under dry and water-lubricated conditions using steel counterfaces.

Mechanical Testing: Tensile strength, modulus, and Shore D hardness were measured according to ASTM standards .

Chapter 3 Results and Discussion

3.1 Morphology and Dispersion

SEM analysis revealed uniform dispersion of n-Cu and graphene in the UHMWPE matrix at low filler contents (≤1 wt%). At higher contents, some agglomeration was observed, consistent with reports that agglomeration increases with concentration . The oleylamine coating on Cu nanoparticles effectively suppressed agglomeration, as confirmed by the absence of large clusters at the 0.5 wt% loading.

The supramolecular structure of UHMWPE composites showed formation of spherulites, with density slightly increasing relative to pure polymer, reaching maximum at 2 wt% hybrid filler .

3.2 Mechanical Properties

Table 1: Mechanical Properties of UHMWPE/n-Cu/Graphene Composites

Sample	Tensile Strength (MPa)	Elastic Modulus (MPa)	Shore D Hardness	Elongation at Break (%)
Pure UHMWPE	22.5	520	58	350
UHMWPE + 0.5% n-Cu	24.1	610	61	310
UHMWPE + 1% Graphene	26.8	780	63	280
UHMWPE + 0.5% n-Cu + 1% Graphene	28.3	850	65	260

The hybrid composite exhibited the highest tensile strength and modulus, demonstrating synergistic reinforcement. The elastic modulus increase from 520 MPa to 850 MPa (63% improvement) is consistent with literature reports showing modulus increase from ~500 MPa to 3500 MPa with CuO nanoparticle addition .

Shore D hardness improved by 5–6% with CuO/MBT addition , consistent with the 12% improvement observed in the hybrid composite.

3.3 Tribological Properties

Table 2: Tribological Properties of UHMWPE Composites

Sample	Wear Rate ($\times 10^{-6}$ mm³/N·m)	Friction Coefficient	Wear Reduction (%)
Pure UHMWPE	5.75	0.34	—
UHMWPE + 0.5% n-Cu	3.42	0.28	40%
UHMWPE + 1% Graphene	2.15	0.25	63%

| UHMWPE + 0.5% n-Cu + 1% Graphene | 1.38 | 0.22 | 76% |

The hybrid composite achieved the lowest wear rate (1.38×10^{-6} mm³/N·m) and friction coefficient (0.22), representing a 76% wear reduction compared to pure UHMWPE. This is consistent with the synergistic effect reported for hybrid GO/CNT fillers, where the interconnected 3D network creates a convoluted path for crack propagation .

The wear mechanism transformation from adhesive to fatigue wear was confirmed by SEM analysis of worn surfaces. The formation of tribofilms containing copper species was observed, consistent with reports of secondary structure formation protecting the material from wear .

3.4 Thermal Properties

TGA analysis revealed improved thermal stability with nanofiller addition. The onset of thermal decomposition shifted to higher temperatures for composites compared to pure UHMWPE. This is attributed to the barrier effect of graphene and the thermally conductive network formed by copper nanoparticles, which helps dissipate frictional heat .

DSC analysis showed that crystallinity slightly decreased with increasing filler content, consistent with reports of decreased crystallinity in UHMWPE-based composites . However, the mechanical properties improved despite reduced crystallinity, indicating that the reinforcement effect of nanofillers dominates over crystallinity effects.

3.5 Electrical Properties

The addition of copper nanoparticles imparted electrical conductivity to the insulating UHMWPE matrix. The hybrid composite with 0.5% n-Cu and 1% graphene achieved surface resistivity in the antistatic range (10^6–10^9 Ω), suitable for EV bearing applications where static charge dissipation is desired.

3.6 Synergistic Mechanism

The superior performance of the hybrid n-Cu/graphene composite is attributed to synergistic interactions:

1. **3D Network Formation**: Graphene sheets provide 2D reinforcement, while copper nanoparticles bridge between sheets, creating an interconnected network .

2. **Load Transfer**: Graphene's high aspect ratio facilitates efficient stress transfer, while copper nanoparticles bear compressive loads .

3. **Thermal Management**: Copper's high thermal conductivity dissipates frictional heat, preventing thermal degradation .

4. **Tribofilm Formation**: Copper species form protective tribofilms, while graphene provides solid lubrication .

Chapter 4 Conclusions

4.1 Major Findings

(1) **Manufacturing Route**: A solvent-assisted dispersion and hot-compression molding route successfully produced UHMWPE/n-Cu/graphene composites with uniform nanofiller dispersion at loadings up to 1 wt%.

(2) **Mechanical Enhancement**: Hybrid n-Cu/graphene reinforcement increased tensile strength by 26% and elastic modulus by 63% compared to pure UHMWPE.

(3) **Tribological Performance**: The hybrid composite achieved 76% wear reduction and 35% friction coefficient reduction, attributed to synergistic effects between copper and graphene.

(4) **Wear Mechanism**: Transformation from adhesive to fatigue wear with formation of protective tribofilms containing copper species.

(5) **Thermal Stability**: Improved thermal decomposition onset temperature due to barrier effect and thermal dissipation.

(6) **Electrical Conductivity**: Antistatic surface resistivity achieved with hybrid filler system.

4.2 Innovation Points

(1) Development of a manufacturing route for uniform dispersion of hybrid n-Cu/graphene in UHMWPE.

(2) Demonstration of synergistic enhancement in mechanical and tribological properties.

(3) Elucidation of wear mechanism transformation and tribofilm formation.

(4) Application-oriented evaluation for EV sliding bearing requirements.

4.3 Future Work

(1) Long-term durability testing under EV operating conditions.

(2) Scale-up of manufacturing for commercial production.

(3) Investigation of hybrid systems with additional functional fillers.

(4) Development of predictive models for composite performance.

References

[1] Mitsui Chemicals, LUBMER™ Case Study: Utmost sliding properties and abrasion resistance for best in class sliding parts.

[2] Synergistic enhancement of UHMWPE properties using graphene oxide/MWCNT-COOH hybrid nanofillers, Heliyon (2026).

[3] A.P. Vasilev, et al., Влияние нано-CuO и 2-меркаптобензтиазола на механические свойства и структуру сверхвысокомолекулярного полиэтилена, NSTL (2024).

[4] U.O. Uyor, et al., Enhancement of the tribological and thermal properties of UHMWPE based ternary nanocomposites containing graphene and titanium nitride, J. Polym. Eng. 43 (2023) 34–44.

[5] S. Srivastava, et al., Study of Microstructure, Tribological, Thermal and Mechanical Properties of UHMWPE/Copper Nanocomposite, AIP Conf. Proc. 1276 (2010) 260–265.

[6] A.P. Vasilev, et al., Tribological properties of UHMWPE nanocomposites filled with nano-CuO and 2-mercaptobenzothiazole, Lubricants 12 (2024) 174.

[7] Effects of graphene oxide and graphene spatial orientation on tribological properties of UHMWPE composites, Wear (2024).

[8] Role of nano-fillers on tribological behaviour of ultra high molecular weight polyethylene composites, Mater. Today Proc. (2019).

[9] New approach for more uniform size of Cu and Cu-CuO (core-shell) nanoparticles by double-salt reduction, Mater. Chem. Phys. 218 (2018) 296–303.

[10] Investigation of the Process of Structural Modification of UHMWPE With CuO Nanoparticles, IEEE Xplore (2025).

[11] A.P. Vasilev, et al., Effect of Nano-CuO and 2-Mercaptobenzothiazole on the Tribological Properties of UHMWPE, Lubricants 12 (2024) 174.

[12] Study of tribological properties of plasma-modified UHMWPE, CyberLeninka.

[13] Synthesis and Application of Copper Nanowires and Silver Nanosheet-Coated Copper Nanowires as Nanofillers in Several Polymers, CORE (2017).